# TECHNOLOGY

In many fictional worlds, the tools and technology found are based on a developmental path that has roots leading back to western European sources. In fantasy, these are often things like broad swords or plated armor. In science-fiction, it is often barreled guns or, well, plated armor. What's more, because of the industrialization in most science fiction settings, technology is thought of as mass manufactured and standardized. There are listings for things that read "The XR-22 is the most common handgun and is used by police forces throughout the City." The weapon has a specific official look and the one your Character buys in a shop will be the same as the one they get in any other shop, because it came off of a factory assembly line.

However, in Coyote & Crow, corporations and industrialization haven't really taken a foothold, at least not to the same degree. Instead, blueprints or recipes, commonly referred to as woyi, are shared for basic items, printed by gats at a local level as needed. Then, technicians and craftsmen customize the item to their own specifications or to the needs of their customer. In this sense, items in Coyote & Crow, while being more technologically advanced than our real world, are closer in feel to a fantasy setting, where someone might commission a blacksmith to make them a sword. The game-related stats for the sword might be exactly the same as from a sword a thousand miles away, but the actual weapon would likely be very distinct.

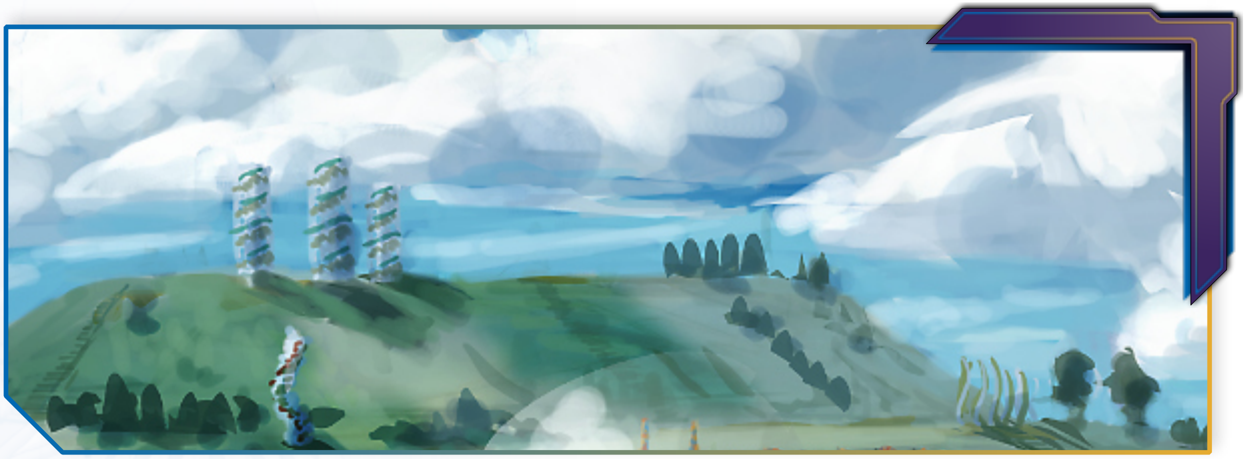
Below are some key pieces of technology that are used in Cahokia and the world over. You'll also find more detail and game-specific descriptions in the Equipment section. It is important to remember that these are broad generalizations and because of the lack of mass production, lots of exceptions and variations exist on everything in this chapter.

## Solar Power

Solar panels are small and similar to reptilian scales in their presentation. Each panel is interconnected to its neighbors. They can be created in any color and often coat building exteriors and vehicle surfaces. Most people don't even notice they are there. These can be printed using commonly available woyi. In larger cities, there are even machines that will "paint" the scales, in any color or pattern, right onto structures or vehicles. This allows for ornate art that also generates power. Solar, despite the cloudy skies, is extremely prevalent and highly efficient. Excess or unused power is diverted to high capacity batteries.

## Wind and Hydroelectric Power

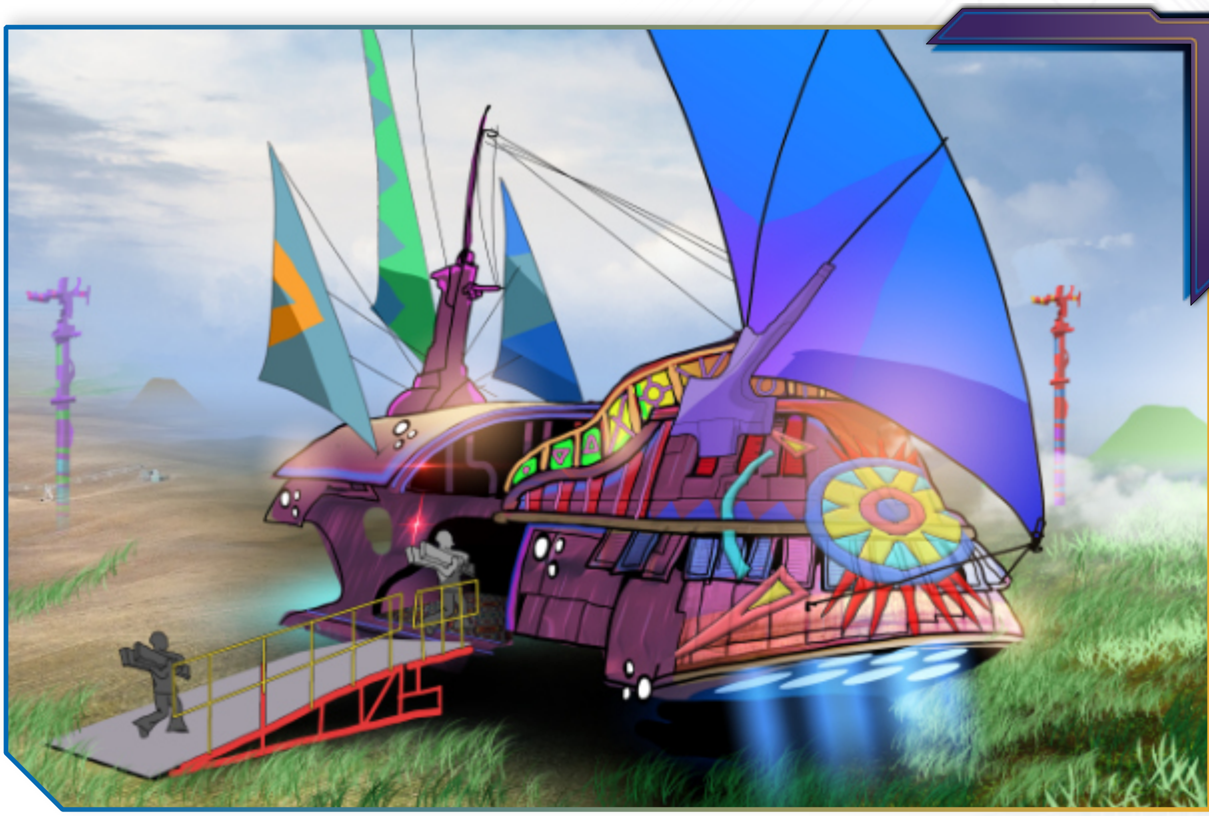
Wind and hydroelectric power engines are also very efficient and deliver clean power directly to users or to batteries. Massive, narrow wind turbines stand atop most homes. Each is custom painted to show various images and patterns, depending on wind speed. Wind turbines tend to be massive, artistic structures while hydro-electric units are often small and portable, meant to be dropped into nearby rivers and streams.



## Yutsu Lift Technology

This technology powers most of the vehicles in Makasing and many around the world. It works by applying an electromagnetic field to Adanadi tissue suspended in a vacuumed ring, which then emits negative gravitons. Attached to the bottom of a vehicle, it allows for it to hover and maintain a constant elevation by pushing off from the ground below it. Larger engines found on barges are able to lift heavy cargo vessels high enough that they can clear tree tops. Yutsu is a common slang word used to describe a number of vehicles that use the technology and is a catch-all term for whatever vehicle someone might mean in the context of their sentence. It might refer to a train, a sled, a personal mobility vehicle (or PMV) for the disabled, a barge, or any number of other vehicles whose primary power is yutsu tech.

Yutsu technology is also responsible for powering mag-bows, modern plumbing, and a host of other things that benefit from a reduction in gravity and therefore weight or drag.



## Second Eyes

To understand Second Eyes, and the important role they play in modern Cahokian society, you have to understand how quickly technology progressed over the previous 100 years in this world. It took our real world 120 years to go from the invention of radio to the ubiquitous use of laptops and wireless data transmission. But think how much of that adaptation happened in the last 20 of those 120 years. In this world, different kinds of evolutionary pressures, combined with the power of the Adanadi, made that same change happen in less than half that time.

While verbal transmission, hand writing on paper, and digital documents are still very commonplace, so are Second Eyes, thanks to 3D printers. Second Eyes are slim-fitting goggles with dark lenses. These goggles at first were hardwired into cable information systems that transmitted news, stories, music, recorded theater performances, and most importantly, spiritual content. These goggles are meant to convey a personal experience to the wearer; using them is seen as a private matter, not done as something social. Watching something on Second Eyes wouldn’t be done in a group setting. While you might watch a recorded theater performance, you wouldn't do it with someone sitting next to you, whether they were watching that same thing on their goggles or not. It would be disrespectful to use the goggles fully immersed with anyone in the room with you.

On the other hand, Second Eyes also have an augmented reality (AR) setting. These allow the goggles to become transparent and superimpose data over the wearer's vision. Younger people are more likely to do this, as are people in tech or military careers who need vital, up-to-date information. There are some worries that younger people are spending too much time isolating themselves from their community and culture using Second Eyes.

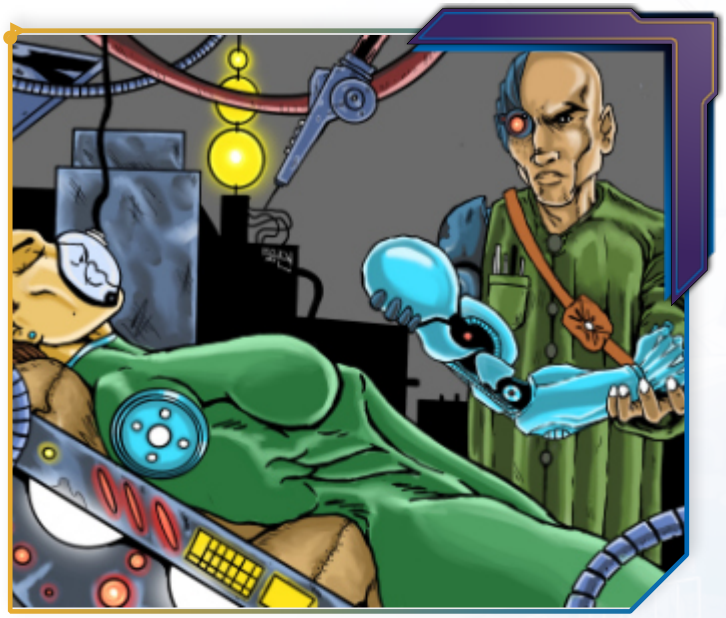
There are fully immersive haptic versions of Second Eyes that truly capture the virtual reality experience. However, these are still fairly uncommon and there are not yet a wide variety of applications.

One of the big differences between the sights and sounds provided by Second Eyes and our real-world version of virtual reality is that, outside of recordings of actual events like theater, there is no desire or inclination to try to make things more realistic. Indeed, many of the visionary designers of Second Eyes environments, interfaces, and entertainment intentionally make the experience surreal and as far away from reality as possible. This is part of the reason that Second Eyes lend themselves well to religious experiences.

## Gats

Gats changed both continents more than any other single invention and are analogous to 3D printers in our world. A number of tribes and nations claim that specific people from their regions were the original inventors of the device; the truth is that the core discoveries came from the Ti'Swaq Alliance. The sharing of that knowledge, while often controversial among politicians, helped millions escape poverty and fight back against the elements. The printers vary in size from a paper printer to those that can print entire walls at a time. 3D printers and their operations are very important and the technicians who operate them are highly trained and maintain an elevated status in their local societies.

The printers generally use biofuels for materials, mahiz for the most part. They are often solar powered, but run on everything from hydro-electric and wind to batteries and even hand cranks. Each of these devices is essentially a small factory and took the place of classical modern industrialization seen in our world. It gave small towns the ability to build durable, inexpensive homes from nothing more than food waste. It allowed cities to massively modernize and upgrade their infrastructures. Gats



also allowed for new technologies to be replicated easily and rapidly.

While basic materials can be replicated simply, more complex materials, including those required to build additional gats, require more than just food waste. Still, the raw mineral and chemical components are relatively easy to obtain, and modern industrial gats themselves can often break down and reconfigure complex components into the materials they need. The technology continues to improve too. The latest gats can create weapons lighter and stronger than titanium that are also biodegradable.

There are two important things to note about these devices. First, while gats helped raise the standard of living across the world and are truly incredible, they haven't really changed how people work or go about their lives. Carpenters are still carpenters. Engineers are still engineers. Farmers still farm. The people of this world see these printers as tools, not an end to itself. Second, most standard objects that don't have complex digital or electronic components are, by default, made to be biodegradable over a few hundred years. This is partially since they are made mostly from biofuels, but it also comes from a reluctance to build landfills, which are few and far between.

Just as important as the printers themselves are the woyi, or blueprints, for various items. These are often stored locally on the printers, like software. They are created, customized, and traded, but rarely bought and sold. Woyi are treated like a seed might be. It has little value in itself outside of a skilled farmer. Instead, it is the craftsmen who bring the final items to life that are considered valuable. The charge for the creation of almost any product is primarily based on the amount of labor necessary to create it, along with some costs for base materials. Rarely is the woyi part of that equation.

## Niisi

Niisi are ubiquitous throughout Cahokia and most urban centers. They are personal computer devices that people wear on their forearms and often span the area from just above the wrist to just shy of the elbow. Frequently, they are customized to match tribal or personal styles. The devices act much the way our laptops and mobile phones do. Niisis can take photographs, store and transmit video and audio, and operate software. Niisis take their name from a word that translates to "grandmother," which is an affectionate, generic nickname that covers all the styles and models of the device. It got this name when niisi became hooked into the Free Lands data network, which put vast amounts of historical and cultural data at the average citizen's fingertips, which some folk equate to having an elder at their side.

Some older folks either eschew niisi or only wear them when necessary, while a minority of younger people see them as a controlling tool of shadowy government forces and avoid them. However, most people use them on a daily basis for communication, business, and scheduling. It is also good to note here that digital photography as an art form doesn't really exist. While still image recording with niisi is possible, it is mostly done for record keeping and other practical issues.

People who do not have or use niisi can easily find public terminals throughout Cahokia. There are also home versions of the device, similar to our real world voice-activated and web-connected assistants.

## Telecommunications

As mentioned above, most people use niisi for direct communication. It should be noted that the transmissions to and from these devices are routed through a series of below-ground cables and above-ground towers. No satellites are in orbit.

Additionally, people living in the Free Lands cannot communicate outside of the Free Lands on a whim. Communications across national borders are tightly controlled by governments, who have strong reasons to be suspect of electronic espionage. Communication lines across borders are limited to specific geographic points and are kept secret and guarded. Additionally, a person from one nation can't go into another nation and start using their information net (see daso, below). Their niisi wouldn't connect properly. This means the device, once out of range of its home network, can not effectively communicate back to its origin nation without serious technological assistance of some kind.

Over major cities like Cahokia, there are a number of dirigible transmission hubs floating at about 20,000 feet. Between these, land towers, and cable systems, data finds its way to where it needs to go within a nation in most cases. These cables, towers, and systems are free to citizens and are considered public access.

## Daso

The daso is the common term for what people in our world call the World Wide Web or the internet. It is distinct in many important ways though. Firstly, while people generically call any net space the daso, what they are usually referring to are a tightly controlled series of intranets. While the technology for digital sharing of information is commonplace, the networks for sharing that information is not. Coming on the heels of the All Tribes War, Cahokia and most other nations are very wary of allowing information to flow easily out past their borders. Instead, there are a series of localized intranets, either government controlled or private, that people can sign onto. As mentioned in the telecommunications section, the information is either transmitted by ground cables, towers, or high-altitude dirigibles. Cahokia has the largest network and can be accessed throughout most of the Free Lands. Trying to connect beyond those borders requires both a landline and a high-security access — or extremely good hacking skills.

Cahokia's daso is free to all citizens and is used for things like voting, city news, private advertising, chat and social platforms, event planning and information, educational broadcasts, and access to private companies that provide recorded arts and entertainment beyond Cahokian borders. Privately owned daso offer a much broader range of content at a cost; not all of it savory.

Hacking into local intranets and servers is illegal but common. For this reason, daso security specialists are increasing in demand as zagabaan, or hackers, are continuing to find ways into secure digital spaces, disrupting business, stealing information and data, and generally causing chaos.

## Space

For most of their history, the people of Coyote & Crow had a very clear physical limit on how high an altitude they could physically ascend to since they couldn't fly. With the advent of gliding and then propulsive flying, things changed rapidly. Wars propelled that technology even farther.

However, for a mix of historical, cultural, and technological reasons, most people, including Cahokians, have not pushed to fly or travel higher than about 30,000 feet. The largest culprit is a combination of societal will and the internal infrastructure necessary to test and develop the technology. For some, there just doesn't seem to be a reason to go higher. For others, there is a deep seated fear in going beyond the limits of the sky. Fears about what happened to the Great Spirit, why the world changed, and what place people have in this world have led some to want to stay close to home as sort of a cultural trauma.

That doesn't mean no one has tried. Solitary scientists and adventurers may have pushed that envelope. Many whisper that various nations, including the Free Lands, are trying to reach higher in secret, hoping to gain a surprise advantage over potential enemies in case of war. As of yet, no nation has claimed an intentional desire to go into space.

## A Further Note on Technology That Doesn't Exist But Could

The technological breakthroughs that exist in Coyote & Crow happened in a similar way as in our world. They just happened in a different order and at a different pace. Much like a tech tree in a video game, some things can be skipped or diverted around. We took one direction, the people of Makasing took another.

A classic example is the wheel. The wheel exists in Coyote & Crow and is often seen in devices in Azayang, where it was in common use even before the Awis. But in Makasing, the wheel never took hold as a common device for vehicles. It wasn't necessarily needed for a number of reasons, like sea and river travel, and technology accelerated so quickly that science and culture simply bypassed it as the basis for further developments. Instead, they have things like yutsu technology, turbofans, magnetohydrodynamic propulsion and the myriad of ways people use waterways to get around, circumventing the need for wheels.

In addition, there isn't really an analog for film and television in Coyote & Crow. Part of that is because in our real world, there was a chain of interrelated discoveries in Europe involving everything from horses to gunpowder that fueled the inventions of photographic plates, film, flashes, and a host of other photography-related technology. Because of this, or rather the lack of this, film and television were never invented. The idea of a flat rectangle where images are conveyed does exist, but only for communication of information, not entertainment, news, or story conveyance. The closest analogy to the cultural impact of these forms of media are Second Eyes.

There are other items that could exist within this world, and indeed, may have already been discovered, possibly more than once, but haven't taken hold. The items mentioned here are specifically in relation to inventions and ideas that are common in our own world or in other popular eurocentric science fiction.

- Nuclear Power and Weapons
- Nanotechnology
- Large Scale Biological Weapons
- Large Scale Chemical Weapons
- Large Scale Explosives For Warfare
- Artillery
- Ballistic Missiles
- Fossil Fuel Production or Use
- Orbital Satellites
- Rockets

That is not to say these things could not take hold or play a part in the future. Maybe the implications or development of one of these might be part of your group's Saga. But for a myriad of historical, sociological, and cultural reasons, these things and others were never developed or widely implemented.

If you are unsure whether or not something exists in this world, use the resources in this book and then discuss with your play group to decide if a particular piece of technology makes sense to include in your story. If so, decide how widespread its use is, how commonplace it is, and how recently it was developed.

## How The Moon Got Its Dents

Coyote loved to play with the animals; most of all he loved to kick ball. All day long, they would play under the sun, until the very last of twilight would seep away, and the dark night would settle under the low, low moon. As the animals would head to sleep, Coyote would cry and cry to keep playing, but they could not see the ball in the dark, and did not want to get hurt.

So Coyote sat in the dark, waiting for the sun to return. Oh, how he wished and wished there was a way to see the ball at night, so he and the animals could play as long as they wanted. Suddenly, as he stared up at the bright moon, Coyote had an idea!

Deep in the forest, Coyote knew of a tree taller than any tree before. He went to it, and began to climb. He climbed and climbed, until he reached the very top. Then, he waited, until sure enough — the low moon, full and bright, passed right over the tree. Coyote leaped from the top of the tree — and he grabbed the moon! He cackled in his cleverness, for he now had the perfect ball to play with the animals. He quickly woke them up, and in their amazement at this new bright ball, they all began to play again. They played and played, kicking the moon around, laughing in delight.

Soon enough, Great Spirit awoke from her rest. But she was startled when she did not see the sun's rays, and even more when she did not see the moon's glow. How would the hunters hunt without morning light? The weavers see their work? The people live? She began to search for the moon, checking the heavens and then moving lower, until she stumbled upon the clearing where Coyote and the animals were playing ball.

***Coyote***, Great Spirit said. ***You took the moon from the sky! The day cannot come until the moon is put away. Give it to me, that I may right your wrong.*** And though Coyote cried for his cleverness to be halted, he did as Great Spirit asked. And Great Spirit wasted no time in placing the moon back into the sky — this time, so very high that even She would have trouble getting it down.

But to this day, you can still see the marks from their game upon the surface! All the kicks the animals did give. What a great game it must have been, for in the lonely night you can still hear Coyote cry for the moon to sink low again.